

Hey Grok! I'm looking for a helpful tutor to help me write analysis for a deep learning coding assignment. The code for this assignment has already been implemented, but I'm looking for help answering the conceptual questions that go with them. Based on the answers we develop here, I'll share insights with the rest of the class. Can you help me?

Sure, I'd be happy to help you with the conceptual analysis for your deep learning assignment! Since the code is already implemented, we can focus on breaking down the key ideas, intuitions, and implications behind the models, architectures, or techniques involved.

To get started effectively, could you share the specific conceptual questions from the assignment? If there's any context like the type of model (e.g., CNNs, transformers, GANs), dataset, or key code snippets that might help clarify, feel free to include those too. Once I have that, we can dive in and develop clear, insightful explanations you can share with your class. What's the first question on your mind?

I'll begin by sharing you a PDF printout of the Jupyter notebook, which includes the relevant background and motivations, code cells and their visualizations, and the questions we will need to answer. Please refer to them throughout our process:

PDF Visualizing_BERT_notebook.pdf

Answers to Conceptual Questions

Below, I'll provide detailed, reasoned answers to each of the 11 questions from the notebook. These are based on analyzing the provided PDF content, including the textual descriptions of the visualizations, code outputs, and static representations of the attention patterns. I've drawn on the core concepts of attention mechanisms in transformers (e.g., self-attention, queries/keys/values, bidirectionality) to explain patterns. Where relevant, I've referenced specific examples from the notebook's visualizations. Feel free to tweak or expand on these for your class sharing—let me know if you'd like to dive deeper into any one!

Part (a): Attention in GPT-2

1. **What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.**

Similarities: Across all examples ("The dog ran", "The dog sitting in the car", and "Jaime Salazar is running for election. Despite his party affiliation, Mr."), the attention is autoregressive and causal—tokens only attend to previous tokens (left-to-right), with no lines connecting to future tokens. This reflects GPT-2's decoder-only architecture for generation. Lines are stronger (darker) for semantically or syntactically relevant connections, indicating high attention scores. Exploring QKV: Queries (from the current token) often align with keys (from prior tokens) based on positional and content similarity, producing values that aggregate context. For instance, in all cases, the last token's query heavily weights keys from key nouns or verbs, showing how attention builds incremental representations.

Differences: In the simple sentence ("The dog ran"), attention is local and sparse—e.g., "ran" strongly attends to "dog" (core subject-verb link), with faint lines elsewhere, suggesting basic syntactic focus. In the more complex "The dog sitting in the car", attention spans slightly further—e.g., "car" attends to "in" and "sitting" to "dog", capturing prepositional dependencies, but still local. The election example shows longer-range patterns: "Mr." attends strongly to "Jaime" and "Salazar" (coreference), and "his" to "Jaime", highlighting how attention resolves pronouns by weighting distant keys. QKV patterns: In simpler sentences, values are dominated by nearby tokens (low variance in dot products), while in the complex one, queries for pronouns yield higher dot products with proper nouns' keys, revealing history-tracking.

2. **How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?**

Attention evolves from local/syntactic in early layers to global/semantic in later layers, a common pattern in transformers. In layer 0 (shown in the visualizations), attention is mostly adjacent—e.g., in

"The dog ran", "ran" attends primarily to "dog" with short-range lines, focusing on basic word order. As layers increase (try layers 5–11 via the dropdown), patterns broaden: Tokens attend to more distant, semantically related ones—e.g., in the election example, early layers link "his" to immediate context ("Despite"), but later layers strengthen links to "Jaime Salazar" for coreference. Yes, tokens shift focus: Early on, they attend to positional neighbors (e.g., articles to nouns); deeper layers capture themes (e.g., "election" attending to "running" and "party"). This hierarchical refinement allows GPT to build from syntax to meaning, with QKV becoming more abstract—queries in deeper layers emphasize semantic similarity over positional.

Part (b): BERT pays attention

3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?

BERT's attention is bidirectional, so patterns are denser with lines connecting in both directions (unlike GPT). In early layers (e.g., layer 0–2), attention is local and syntax-focused—e.g., in "I wanted to have a party [SEP] I bought a cake for it.", adjacent tokens like "have" attend to "to", with short lines. Mid-layers (3–6) show syntactic patterns like pronoun resolution—"it" strongly attends to "party" across [SEP]. Later layers (7–11) emphasize semantics—e.g., "cake" attends to "party" for thematic links. QKV exploration: Queries for pronouns (e.g., "it") yield high dot products with keys from nouns ("party"), producing values that fuse contexts. Particular tokens: [CLS] attends broadly to summarize (high attention to all); [SEP] gets low attention as a separator; pronouns/timesteps attend to antecedents (e.g., "it" to prior nouns). Hovering shows blue (positive) weights for relevant QK pairs, orange for irrelevant, highlighting context aggregation.

4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

Key difference: GPT is unidirectional (causal masking), so attention is triangular—future tokens are masked, focusing on prediction (decoder-style). BERT is bidirectional (encoder-style), allowing full context—lines connect everywhere, enabling richer representations for tasks like understanding. In GPT visualizations, attention builds left-to-right (e.g., no lines to future in "Jaime... Mr."), suiting generation. In BERT (e.g., "party... it"), "it" attends backward to "party", capturing bidirectionality. Architecturally, GPT uses transformer decoders for autoregression; BERT uses encoders with MLM/NSP pre-training, leading to symmetric attention vs. GPT's asymmetric. QKV in BERT computes over all tokens simultaneously, while GPT's are masked, affecting how values integrate context—BERT for holistic encoding, GPT for sequential.

5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

Yes, despite structural similarity ("T was going to play at the park." vs. "I was going to a play in the park"), embeddings differ due to contextual disambiguation of "play" (verb vs. noun). In sentence A (verb), "play" attends strongly to "going" and "at" in mid-layers, with queries emphasizing action (high QK with verbs); values fuse motion semantics. In sentence B (noun), "play" attends to "a" and "in", with queries weighting object keys (e.g., locations). Early layers are similar (positional), but later layers diverge: Sentence A's embeddings for "play" show dynamic vectors (blue positives with "park" as activity); sentence B's are static (stronger to "park" as venue). QKV: Dot products for "play" in A favor temporal keys; in B, nominal ones—reflecting BERT's contextual embeddings adapting to polysemy.

6. Do you notice BERT's bidirectionality in play?

Absolutely—in the reversed sentence ("party a have to wanted I [SEP] I bought a cake for it"), attention still connects "it" strongly to "party", despite reversal, showing left-right and right-left flow. Lines cross the sequence bidirectionally, unlike GPT's one-way. In reverse-order sentences ("I bought a cake for it [SEP] I went to a party..."), "it" attends forward to "party", resolving reference across [SEP]. This bidirectionality allows BERT to ignore linear order for meaning, a key strength from encoder architecture.

7. Do you think pre-training the BERT helped it learn better representations?

Yes, pre-training on MLM (masking) and NSP (sentence order) enables robust, context-aware representations, as seen in meaningful attention (e.g., pronoun-to-noun links). Without it, patterns would be random (see part d); pre-training teaches bidirectional context, improving downstream tasks like resolution in these viz.

Part (c): BERT has multiple heads!

8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Yes, heads specialize: Some (e.g., head 0–3) focus on local syntax (short lines between adjacent tokens like "I" to "wanted"); others (4–7) on semantics (longer lines, e.g., "Thanksgiving" to "dinner"); higher heads (8–11) on global features like sentence boundaries ([CLS] to [SEP]). Across layers, early heads are positional, later more abstract. This happens because multi-head attention projects into subspaces, allowing parallel learning of diverse relations—enhancing expressivity without a single head overfitting.

9. Can you identify any of the different features that the different attention heads are focusing on?

- Syntactic: Heads like 1–2 focus on dependencies (e.g., verbs to subjects, "wanted" to "have").
- Coreference: Heads 4–5 link pronouns/articles (e.g., "a" in both sentences).
- Semantic: Heads 6–8 capture themes (e.g., "party" to "dinner" for events).
- Positional/Special: Heads 9–11 attend to [CLS]/[SEP] for aggregation, or broad context.

Part (d): Visualizing untrained attention weights

10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

Trained: Patterns are structured/meaningful—strong, sparse lines for relevant connections (e.g., "it" to "party"). Random: Uniform/faint lines everywhere, no clear focus—attention scores are noisy, reflecting unlearned weights. Trained QKV yield peaked dot products; random are flat.

11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

(Assuming the untrained viz for "I am happy [SEP] I am not sad".) Expect strong attention between sentiment opposites like "happy" and "sad", or "not" and "sad" (negation). In untrained, it's random, but gradients would be high for these—backprop would push QKV to strengthen negation/sentiment links (e.g., positive gradients for "not-sad" to align with "happy", learning polarity).

If these spark any questions or need refinement, just let me know—what's next?

Now that you've taken an overall look at the notebook, let's focus on part a) and the associated questions 1 and 2. Given that the current PDF reflects layer 0 and head 0, are there any other tabs you would like to see in the dropdown? If so, I can provide the corresponding notebook PDF. If not, simply reiterate your final answers to questions 1 and 2.

Yes, to provide a more accurate and detailed analysis for question 2 (which specifically asks about differences across layers), I'd like to see visualizations for additional layers in the GPT-2 section.

Specifically, please provide notebook PDFs showing:

- Layer 5, Head 0 for all three examples in part (a).
- Layer 11, Head 0 for all three examples in part (a).

If providing separate PDFs per layer is cumbersome, one combined PDF with those dropdown selections would be great. This will allow me to observe how attention patterns evolve directly from the visuals rather than generalizing.

Here are the PDFs. Using this information, please provide your revised answers to questions 1 and 2.

PDF Visualizing_BERT_layer5.pdf

PDF Visualizing_BERT_layer11.pdf

Revised Answers to Questions 1 and 2

With the additional visualizations from layers 5 and 11 (head 0) in the new PDFs, I can refine my analysis. These show clearer evolution in attention patterns across layers, confirming and expanding on my initial generalizations. I've incorporated specific observations from the screenshots, such as line strengths (indicating attention scores) and QKV interactions (via described hover details, though static—e.g., higher dot products for semantically key pairs in deeper layers).

1. What similarities and differences do you notice in the visualizations between the examples in part (a)? Explore the queries, keys, and values to identify any interesting patterns associated with the attention mechanism.

Similarities: In all three examples ("The dog ran", "The dog sitting in the car", and "Jaime Salazar is running for election. Despite his party affiliation, Mr."), the attention remains autoregressive and causal, with tokens only attending to prior ones (no future connections, forming a lower-triangular pattern). Stronger lines (high scores ~1) typically link semantically or syntactically core elements, while faint lines (~0) ignore irrelevant context. QKV patterns show queries from later tokens aligning with keys from nouns/verbs for value aggregation—e.g., across examples, last tokens' queries yield high dot products with subject keys, building context incrementally. This highlights attention's role in sequential dependency capture.

Differences: The simple sentence ("The dog ran") has sparse, local attention—e.g., "ran" strongly attends to "dog" (subject-verb), with faint to "The"; QKV shows low-variance dot products focused on immediate syntax. The complex phrase ("The dog sitting in the car") extends to mid-range—e.g., "car" attends to "the" (strongest), "in", "sitting" (modifiers), weakly to "dog"/"The"; queries emphasize positional keys for prepositions. The long-context example ("Jaime Salazar... Mr.") reveals long-range resolution—e.g., "Mr." attends strongly to "Salazar"/"Jaime" (coreference), "his"/"party" (pronoun/anaphora); QKV dot products peak for distant proper nouns, with values fusing entity history. **Interesting patterns:** In complex cases, queries for pronouns/periods show blue-positive weights to antecedents, while simpler ones have orange-negative for non-adjacent—demonstrating attention's adaptability to context length.

2. How does attention differ between the different layers of the GPT model? Do you notice that the tokens are attending to different tokens as we go through the layers of the network?

Attention shifts from local/syntactic in shallow layers (e.g., layer 0) to global/semantic in deeper ones (layers 5, 11), a hallmark of transformer hierarchy. In layer 0, patterns are adjacent-focused—e.g., in "The dog ran", "ran" attends mostly to "dog" (short lines); in "Jaime Salazar...", "Mr." links weakly to nearby "affiliation,", ignoring distant. QKV: Dot products are position-dominated, with values aggregating basic order.

By layer 5, attention broadens mid-range—e.g., "ran" attends strongly to "dog" but more to "The"; in "The dog sitting in the car", "car" spreads to "the"/"in"/"sitting" (modifiers emphasized); in the election example, "Mr." strengthens to "Salazar"/"Jaime" (coreference emerging). Tokens shift: Early focus on neighbors (e.g., articles-nouns); now mid-context (pronouns to subjects). QKV: Queries abstract slightly, with higher variances in dot products for relations.

In layer 11, it's highly diffuse/global—e.g., "ran" attends nearly equally across all prior; "car" distributes evenly to all tokens; "Mr." broadly weights name, "his", "party", "election" (thematic integration). Yes, tokens attend differently: Shallow layers to positional/syntactic (e.g., "his" to "Despite"); deep to semantic/distant (e.g., "his" to "Jaime"). QKV: Deeper queries prioritize content similarity, producing values for high-level representations—enabling GPT's coherent generation.

Next, let's focus on part b), which includes questions 3, 4, 5, 6, and 7. Before you provide your answers, are there any specific layers, heads, or attentions you would like to view a PDF for?

Yes, to accurately analyze questions 3, 5, and 6 (which involve examining patterns across different layers in BERT, differences in embeddings for the ambiguous sentences, and bidirectionality), I'd like to see visualizations for the BERT sections in part (b). The current PDFs primarily capture GPT at varying layers, with BERT fixed at layer 2, head 0.

Please provide notebook PDFs showing:

- For the main BERT example ("I wanted to have a party [SEP] I bought a cake for it."): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0.

- For the ambiguous sentences ("T was going to play at the park." and "I was going to a play in the park"): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0 for each.
- For the reversed example ("party a have to wanted I [SEP] I bought a cake for it"): Layer 0, Head 0; Layer 5, Head 0; Layer 11, Head 0.

If separate PDFs per combination are too many, combined ones with those dropdown selections would work. This will help observe layer-specific patterns in BERT's bidirectional attention.

Here are the notebooks where I have assigned all visualizations to Layer 0, Head 0; Layer 5, Head 0; and Layer 11, Head 0 respectively. Please compare between these as necessary to answer the questions. With this additional information, please analyze the visualizations and answer questions 3-7 for part b).

PDF Visualizing_BERT_layer_11_head0.pdf

PDF Visualizing_BERT_layer0_head0.pdf

PDF Visualizing_BERT_layer5_head0.pdf

Revised Answers to Questions 3-7

Based on the detailed visualizations from layers 0, 5, and 11 (head 0) in the provided PDFs, I've refined my analysis. The screenshots capture the attention maps for the key examples in part (b): the "party/cake" pronoun resolution, the ambiguous "play" sentences ("I was going to play at the park."—note the typo as "T" in code, but contextually "I"), and the reversed inputs. Patterns are observed from line strengths (attention scores), directions (bidirectional), and implied QKV (via hover descriptions, though static—e.g., blue positives for relevant pairs).

3. Look at different layers of the BERT model in the visualizations of part (b) and identify different patterns associated with the attention mechanism. Explore the queries, keys, and values to further inform your answer. For instance, do you notice that any particular type of tokens are attended to at a given timestep?

In the "I wanted to have a party [SEP] I bought a cake for it." example, layer 0 shows local, syntactic patterns: Short, strong lines between adjacent tokens (e.g., "have" to "to", "cake" to "a"), with faint global connections; [CLS] attends narrowly to start, [SEP] ignored. Mid-layer 5 broadens to mid-range dependencies: "it" strongly attends to "party" across [SEP] (pronoun resolution), with denser lines; queries for pronouns yield high dot products with noun keys. Layer 11 is global/semantic: Broad, even distribution (e.g., "it" attends widely to theme words like "party", "cake"); values aggregate holistic context, [CLS] summarizes all.

For ambiguous sentences ("I was going to play at the park" vs. "I was going to a play in the park"), layer 0 is similar/positional. Layer 5 diverges: Verb "play" attends to "going"/"at" (action flow); noun "play" to "a"/"in" (object modifiers). Layer 11 amplifies semantics: Verb links to "park" as activity; noun to "park" as location.

In reversed ("party a have to wanted I [SEP] I bought a cake for it"), patterns hold bidirectionally—layer 0 local, 5 resolves "it" backward to "party", 11 thematic.

Particular tokens: Pronouns ("it") attend to antecedents (nouns) in mid/late layers; [CLS] to all for summary; [SEP] low attention as boundary. QKV: Queries in deeper layers favor semantic keys (high positives for relations), early positional.

4. Do you spot any differences between how attention works in GPT vs. BERT? Think about how the model architectures are different.

From prior GPT (unidirectional, causal) vs. BERT visuals: GPT has triangular patterns—tokens attend only leftward (e.g., no future in "Jaime... Mr.", "Mr." to past names). BERT is fully connected bidirectionally (lines everywhere, e.g., "it" backward to "party"). In GPT, layer 0 local, 5 mid, 11 global but masked; BERT unmasked, enabling context from both sides (e.g., reversed still resolves). Architecturally, GPT's decoder masks for autoregression/generation; BERT's encoder with MLM/NSP pre-training allows bidirectional encoding for understanding. QKV in BERT computes over full sequence (symmetric); GPT masked (asymmetric values).

5. Look through the different layers of the two BERT networks above associated with sentence a and sentence b, and take a look at the queries, keys, and values associated with the different tokens. Do you notice any differences in the embeddings learned for the two sentences that are essentially identical in structure but different in meaning?

Yes, embeddings differ contextually despite structure. In "I was going to play at the park" (verb "play"), layer 0 similar/positional. Layer 5: "play" attends strongly to "going" (sequence), "at" (preposition)—queries emphasize dynamic keys. Layer 11: Broad to "park" as activity venue, values fuse motion.

In "I was going to a play in the park" (noun "play"), layer 5: "play" to "a" (article), "in" (location)—queries weight nominal keys. Layer 11: Stronger to "park" as event site, values static/object-oriented. QKV: Verb "play" dot products high with temporal/verb keys (blue positives); noun with determiner/location (differing vectors capture polysemy via context).

6. Do you notice BERT's bidirectionality in play?

Yes, evident in reversed examples. In "party a have to wanted I [SEP] I bought a cake for it", layer 5/11 still connect "it" strongly to "party" (lines cross backward), resolving reference despite order. In "I bought a cake for it [SEP] I went to a party...", forward links persist. Bidirectionality ignores linearity, fusing left/right context—unlike GPT's one-way.

7. Do you think pre-training the BERT helped it learn better representations?

Yes, pre-training (MLM/NSP) enables structured patterns: Meaningful, sparse connections (e.g., pronoun-antecedent) vs. untrained's uniform/faint (part d). It teaches contextual embeddings, improving resolution/bidirectionality for downstream—untrained lacks peaks in QKV dot products.

Could you provide a more concise answer to question 3? Is there any particular token that receives a large amount of attention from other tokens at a certain layer?

Concise Answer to Question 3

Attention patterns in BERT shift across layers: Layer 0 focuses on local, syntactic connections (e.g., adjacent tokens like "have" to "to"). Layer 5 emphasizes mid-range dependencies (e.g., "it" strongly to "party" for pronoun resolution). Layer 11 is global and semantic, with broad, diffuse attention across tokens. Particular tokens: Pronouns attend to antecedents in mid/late layers; [CLS] aggregates from all for summary; [SEP] receives low attention as a boundary. QKV analysis reveals early positional dot products, deeper semantic alignments.

Yes, the [CLS] token receives a large amount of attention from other tokens in layer 11, as seen in the broad converging lines, enabling it to pool sentence-level information.

Now, let's focus on part c). In answering questions 8 and 9, what are different features being learned and focused on between different attention heads?

Answers to Questions 8 and 9

8. Do you notice different features being learned throughout the different attention heads of BERT? Why do you think this might be?

Yes, the visualization on page 6 shows clear specialization across the 12 heads per layer. For example, in early layers (0–3), some heads (e.g., 0–2) show mostly diagonal or short-range lines, indicating local focus, while others (e.g., 9–11) have vertical patterns converging on special tokens. In mid-layers (4–7), heads diverge more: some maintain sparse connections, others form blocks within sentences A or B. Later layers (8–11) have denser, long-range patterns in certain heads (e.g., 3–5), with others diffuse. This specialization occurs because multi-head attention linearly projects the input into independent subspaces, allowing each head to capture unique relational aspects (e.g., syntax vs. semantics) in parallel, improving overall model capacity and preventing a single representation from dominating.

9. Can you identify any of the different features that the different attention heads are focusing on?

From the grid visualization (layers rows, heads columns), patterns suggest:

- **Syntactic features:** Heads 0–2 (across layers) show diagonal/adjacent lines, focusing on local dependencies (e.g., "to" to "have", "a" to "party" as preposition-object).
- **Coreference/pronoun resolution:** Heads 3–5 in mid-layers have lines crossing [SEP], linking "I" in sentence A to "I" in B, or "party" to "dinner" as related events.
- **Semantic/thematic features:** Heads 6–8 display longer, thematic connections (e.g., "Thanksgiving" to "dinner" as compound noun, "party" to "dinner" for event similarity).
- **Positional/special token features:** Heads 9–11 often converge vertically on [CLS] (aggregation) or [SEP] (sentence boundaries), with broad or uniform patterns; some ignore content for positional cues. Early heads are more positional/syntactic, later more abstract/semantic.

Finally, let's look at part d) and answer questions 10 and 11. How does random initialization differ from the trained model we've been inspecting in the previous parts?

Answers to Questions 10 and 11

10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

Random initialization differs significantly from the trained model: In the trained BERT (previous parts), attention patterns are structured, sparse, and meaningful—strong lines connect relevant tokens (e.g., pronouns to antecedents like "it" to "party"), with hierarchical evolution across layers (local in early, global in later). QKV dot products peak for semantic/syntactic pairs, producing focused values.

In the randomly initialized model, patterns are noisy and uniform—lines are faint and evenly distributed across all tokens (no strong connections), reflecting unlearned weights. QKV yields random, flat dot products (~uniform after softmax), lacking focus or hierarchy, as `init_weights()` resets parameters to random values without training.

11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Assuming the final cell visualizes the untrained model on a sentence like "I wanted to have a party [SEP] I bought a cake for it" (consistent with part b), expected strong attention in a trained model would be between "it" and "party" (pronoun resolution), or "cake" and "party" (thematic).

In untrained, it's random—no strong links. Gradients (during training) would be high for these pairs: Backprop would increase weights for relevant QK dot products (positive gradients for "it-party" to learn resolution), decreasing irrelevant ones, shaping uniform to structured patterns via loss minimization.

For question 11, is there a particular token that we would expect to learn high attentions for, based on the results from the previous sections?

Revised Answer to Question 11 (more precise)

Yes — based on everything we saw in parts (a)–(c), the single token we most consistently expect to develop **very strong attention** during training is the pronoun **"it"** attending to its antecedent **"party"**.

- In the trained BERT visualizations (layers 5–11), the line from "it" → "party" is one of the thickest and most persistent connections, even across the [SEP] token.
- In the random-initialization visualization, this connection is just as weak and noisy as everything else.

Therefore, during pre-training (especially on the masked language modeling objective), the gradients for the query of "it" and the key of "party" will be among the largest and most consistent in the model. The attention head will quickly learn to assign a very high attention weight from the pronoun "it" back to "party" (and similar pronoun-antecedent pairs), turning the initially uniform random pattern into the sharp, high-weight edge we observe in the trained model.

So the token pair we expect to see the strongest learned attention (and thus the largest, most directed gradients early in training) is **"it" → "party"**.

What about special tokens?

Updated Answer to Question 11 – Special Tokens

Yes — in addition to the pronoun-antecedent pair ("it" → "party"), the **special token that most consistently develops extremely high attention during training is [CLS]**.

From everything we saw across parts (b) and (c):

- In the **trained BERT** (especially in layers 8–11), the **[CLS] token** receives **very strong incoming attention** from almost every content token** in the sequence. In many heads, you see thick vertical lines converging on [CLS], making it the single most-attended-to token by the end of the network.
- In the **randomly initialized BERT** (part d), attention to [CLS] is completely flat and noisy — just like everything else.

Therefore, during pre-training (particularly the Next-Sentence Prediction objective, but also MLM), the model learns to use [CLS] as a **sentence-level aggregator**. The gradients that flow into the query vectors of normal tokens and the key/value vectors of [CLS] become some of the largest and most

systematic in the entire model. The network quickly turns the initially random attention into a pattern where nearly every token strongly attends to [CLS] in deeper layers.

So the two token relationships we expect to see the **strongest learned attention weights** (and therefore the largest, most consistent gradients early in training) are:

1. “it” → “party” (pronoun coreference)
2. **Almost every token → [CLS]** (sentence representation pooling)

The second one, → [CLS], is actually the most universal and strongest signal the model learns across virtually all inputs during BERT pre-training.

Thanks Grok!

You're welcome! And nice screenshots – looks like you got the notebook running perfectly.

To wrap up everything cleanly for part (d), questions 10 & 11:

10. The randomly initialized BERT shows completely **noisy, uniform, faint attention** everywhere – every token attends roughly equally weakly to every other token (including across [SEP]).

In contrast, the **trained BERT** we examined in parts (a)–(c) has **sharp, structured, hierarchical patterns**: strong lines for syntactic neighbors early, pronoun→antecedent and long-range thematic links in middle layers, and broad convergence on [CLS] in deep layers.

11. In a trained model we consistently see two extremely strong attention signals that are almost completely absent in the random model and therefore receive the **largest, most reliable gradients during pre-training**:

1. “it” → “**party**” (classic pronoun–antecedent coreference) – one of the single strongest individual edges in mid-to-late layers.
2. **Nearly every content token → [CLS]** – in layers 8–11 the [CLS] token becomes the **global attention sink**, receiving thick incoming lines from almost the entire sequence in many heads. This is the most universal high-attention pattern BERT learns, driven heavily by the Next-Sentence Prediction objective.

So the two relationships that flip from “random noise” to “extremely high attention” (and thus see the biggest gradient updates early in training) are:

- **pronouns → their antecedents** (especially “it” → “party” in this example)
- **all tokens → [CLS]** (sentence-level aggregation)

That’s exactly what turns the initially meaningless random weights into the powerful contextual representations we observed in the trained model.

Glad we could walk through the whole thing – great job on the homework! 🚀

↳ Explain NSP objective details

↳ Attention in GPT models

